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An AI-Powered Product Identity Form Design Method Based on Shape Grammar and Kansei Engineering: Integrating Midjourney and Grey-AHP-QFD

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Abstract: Product Identity (PI) is a strategic instrument for enterprises to forge brand strength through New Product Development (NPD). Concurrently, facing increasingly fierce market competition, the NPD for consumer emotional requirements (CRs) has become a significant objective in enterprise research and development (R&D). The design of new product forms must ensure the continuity of PI and concurrently address the emotional needs of users. It demands a high level of experience from designers and significant investment in R&D. To solve this problem, a generative and quantitative design method powered by AI, based on Shape Grammar (SG) and Kansei Engineering (KE), is proposed. The specific method is as follows: Firstly, representative products for Morphological Analysis (MA) are selected, SG is applied to establish initial shapes and transformation rules, and prompts are input into Midjourney. This process generates conceptual sketches and iteratively refines them, resulting in a set of conceptual sketches that preserve the PI. Secondly, a web crawler mines online reviews to extract Kansei words. Factor Analysis (FA) clusters them into Kansei factors, and the Grey Analytic Hierarchy Process (G-AHP) calculates their grey weights. Thirdly, after analyzing the PI conceptual sketches for feature extraction, the features are integrated with CRs into the Quality Function Deployment (QFD) matrix. Experts evaluate the relationships using interval grey numbers, calculating the optimal ranking of PI Engineering Characteristics (PIECs). Finally, professional designers refine the selected sketches into 3D models and detailed designs. Using a Chinese brand as a case study, we have designed a female electric moped (E-moped) to fit the PI and users' emotional needs. Through a questionnaire survey on the design scheme, we argue that the proposed innovative method is efficient, applicable, and effective in balancing the product form design of PI and user emotions.

Keywords: product form design; PI; SG; G-KE; G-QFD; Midjourney; E-moped

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1. Introduction

Product identity (PI), also known as Product Identity Recognition, allows enterprises to establish brand value through PI design [1]. Small- and medium-sized enterprises (SEMs) are increasingly focusing on retaining PI elements in product form design to forge a stronger connection with their brand strategy [2,3]. However, focusing solely on brand value expression makes it difficult for enterprises to achieve commercial success [4]. In the era of the experience economy, users place greater emphasis on the relationship between emotional expression and product form. Enterprises need to integrate user emotions into the product form design [5]. Therefore, product form design that retains PI design characteristics while also meeting the emotional needs of users has become a challenge for SMEs in the NPD process. So far, scholars have rarely formed systematic research achievements regarding this issue.

Shape Grammar (SG) has been widely applied in architectural design, engineering design, and graphic design. In the field of product design, SG can maintain the consistency of PI [6–8]. Applying SG in product design faces operational difficulties with parametric encoding and variation rules for shape transformation. Parametric modifications demand a high level of proficiency in both the computer and mathematical fields, while form design through variation rules relies entirely on the designer's professional experience [9]. However, influenced by the designer's experience and talent, the outcomes are often uncontrollable. With the emergence of AIGC, scholars have begun to explore the feasibility of AI-Assisted Design (AIAD) [10].

Kansei Engineering (KE) is a technique that quantifies users' emotional needs and calculates them using mathematical methods, which effectively captures the users' emotional needs and enhances user satisfaction [11]. Since users' emotional needs are subjective and difficult to express, scholars adopt quantitative methods such as questionnaire interviews, user interviews, and survey research to objectively explore user emotional information [12]. As AI technology becomes more prevalent, web crawler tools efficiently gather extensive online data, overcoming the subjectivity associated with small sample sizes [13]. Furthermore, quantifying and analyzing the obtained users' emotional needs will guide the next steps in NPD.

Quality Function Deployment (QFD) is a design research method that can transform Customer Requirements (CRs) into product Engineering Characteristics (ECs) [14]. Its core involves inputting CRs on the left wall and entering product ECs on the ceiling of the House of Quality (HoQ) to construct a correlation matrix. Inviting experts to score the correlations will ultimately calculate the priority ranking of ECs that meet CRs [15]. In the traditional QFD method, correlation is represented by a triangle (▲) for 1 point, a circle (○) for 2 points, and so on [16]. In SMEs, there are relatively few experts, and subjective judgments can introduce uncertainties. Grey set theory is often used to address issues where the sample size is small and the information is unclear [17].

In summary, SMEs lack efficient product design methods that have both established PI and meet user emotional needs during NPD. They also lack the involvement of a large number of experts in R&D and a systematic approach to scientifically address fuzzy and implicit information. More accurately, this study aims to address the following issues:

Q1: How can AIGC technology be effectively utilized to help SMEs quickly obtain a large number of PI concept sketches?

Q2: How can SMEs be helped to quickly obtain and accurately filter customer emotional requirements?

Q3: How can SMEs be helped to integrate CR information when screening PI concept sketches?

To address these issues, this study proposes an AI-powered method that integrates SG and KE to guide SMEs in new product development. Firstly, SG and Midjourney are used to rapidly generate a multitude of PI conceptual sketches, greatly enhancing the efficiency of the creative stage. Secondly, KE and web crawling techniques are used to obtain a comprehensive understanding of user emotional preferences, assisting enterprises in accurately acquiring CRs. Thirdly, the QFD framework is used to construct the mapping relationship between CRs and PIECs. Experts are invited to score the correlations using interval grey numbers to screen out the PIECs that can most effectively meet CRs. Finally, the design scheme is completed by the selected PIECs. The proposed method helps SMEs quickly obtain product form design schemes that meet PI and user performance with less resource investment, enhancing the competitiveness of the enterprise's brand and products.

2. Literature Review

2.1. SG and Midjourney

SG is a design method based on "shape" variations proposed by Stiny and Gips in 1972. It has been widely applied in the fields of architectural design, engineering design, and product design [18]. Essentially, shape grammar is a technique that uses rules to

redesign or modify an initial shape [19]. In product design, some scholars applied SG to the parametric design of a coffee machine, creating 100 handcrafted rules and labels. They implemented a Java application to generate innovative design concepts, demonstrating the potential of SG in product innovation [20]. SG was also used to bridge product branding with form design, as exemplified by Harley-Davidson Motorcycles, encoding 45 shape rules to validate the method's feasibility for PI consistency and engineering applications [21]. Other scholars analyzed over six decades of Buick's front design, using SG to encode and establish 63 rules, effectively validating SG's capability for innovation in form design while maintaining PI recognition [22].

In today's era of AI, AIAD has become a possibility, significantly accelerating the speed of generative design schemes and enhancing the convenience of human-computer interaction. Midjourney and ChatGPT were used to quickly generate numerous color schemes for products, exploring the use of AIGC to enhance color-matching design [23]. DALL-E 2, Midjourney, and StableDi platforms were explored in the field of architectural design, combined with the use of Natural Language Processing to analyze typical usage patterns, and the results suggested that architects and designers can greatly enhance their creativity in the conceptual design phase by using AI tools [24]. From the above studies, SG can effectively maintain PI elements in product design but requires extensive expertise for complex parameter coding. Although some scholars have used Rhinoceros 5.0 software for processing, complex computer and math skills are still needed. AIGC has shown promise in product design research, yet few have explored its integration with SG for PI design.

In summary, the combined use of SG and Midjourney may present a great opportunity to help SMEs quickly obtain a large number of PI concept sketches.

2.2. KE

KE studies users' sensations and impressions, aiming to effectively measure people's psychological perceptions of products and translate them into design characteristics. It was initially introduced by the Japanese scholar Nagamachi [25]. Currently, KE research mainly focuses on two directions. In the first one, during the user-oriented innovative design process, KE tools are used to measure users' psychology, and the precise measurement of users' perceptions guides the design. Some scholars conducted a form design study on ear thermometers, employing a combination of KE and AHP to measure and make decisions regarding users' emotional needs. Then, they established a mapping model between requirements and alternative solutions, selected the optimal scheme for detailed design, and effectively achieved a design that meets users' emotional demands [5]. Other scholars effectively captured users' emotional needs using an improved KE method and combined it with the Kano model for the innovative design of sustainable service, demonstrating the effectiveness of KE in capturing and applying user needs in service design innovation [26]. In the second one, during the product development process, KE is adopted as an indicator that influences user satisfaction. By combining it with other innovative methods in R&D, the efficiency of product development is improved. KE and knowledge graph were proposed as a research framework, which mines consumers' emotional needs and constructs a knowledge management model oriented towards these needs. Then, it identified the knowledge with mapping relationships in professional designers' knowledge systems and finally conducted an effectiveness study with chair design knowledge as an example [27]. KE and TRIZ were proposed as an integrated method for researching the recycling of ecologically innovative products, which has not only met user needs but also achieved the goal of environmental protection, fully demonstrating the effectiveness of KE in capturing CRs during product development [28]. In recent years, with the development of internet technology, traditional questionnaires and survey methods have had limitations due to the small data they collect. Many scholars have started to research the comprehensiveness and feasibility of user reviews and data mining in CR surveys. A data mining technique was developed by using online reviews to categorize consumer emotions and applied heuristic

deep learning to KE, successfully translating extensive user emotional data into design parameters and improving the objectivity of design results [29].

In summary, to help SMEs quickly obtain and accurately filter customer emotional requirements, KE, as a user-centered approach, can effectively capture consumers' emotions and perceptions, transforming them into design elements. During product development, this method requires the integration of data mining, engineering management, and mathematical algorithms to objectively measure and accurately prioritize user emotions, assisting enterprises in identifying key emotional needs. At the same time, CRs need to form a quantifiable mapping relationship with design elements to accurately pinpoint design objectives.

2.3. QFD-Integrated Technologies

QFD originated in Mitsubishi's Kobe shipyard in 1972, which is a process that helps businesses translate real CRs into product features, engineering characteristics, and production details, ultimately creating products that meet customer satisfaction [30]. QFD can effectively enhance design quality, reduce engineering time, optimize design and manufacturing, and decrease the number of product components. It typically includes four interrelated processes that translate CRs into technical characteristics, product features, and manufacturing processes, with the HoQ being the core for mapping the relationship between CRs and ECs [31]. By analyzing and quantifying the correlation between CRs and ECs, the results are input into the HoQ for calculation and ranking, which then guide the product design and development based on the prioritized ECs [32]. The key elements in the HoQ are customer requirements, engineering characteristics, the correlation matrix, competitive analysis, and the importance ratings of CRs and ECs [33]. Experts rate the correlation of CRs and ECs using multi-criteria decision analysis, marking it in the HoQ for calculation [34] (Figure 1).

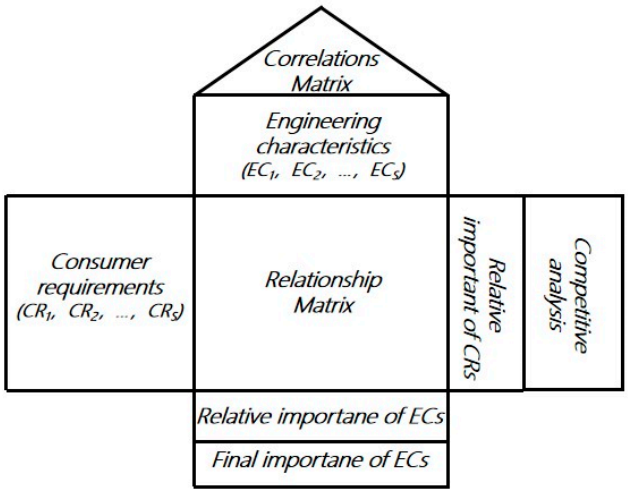


Figure 1. Components of the HoQ.

Recently, scholars have begun to explore the combined application of QFD with other methods, thereby optimizing the application scenarios of this method and enhancing efficiency. QFD method was applied in a scenario-driven approach for the sustainable design of products and service systems, effectively capturing the dual-layer requirements of medical experts and elderly users through the study of nursing beds, and proposed a more accurate functional requirement solution [35]. Radical Innovations (RIs) were introduced into the framework of QFD and TRIZ, supporting the RIs process by extending it across two stages of the HoQ method. The effectiveness of this innovative approach was validated by using the example of a wind power generation system [36]. The Kano model and the TOPSIS algorithm were integrated into the QFD model to measure uncertainties and behavioral risk factors in Chinese e-commerce service systems, demonstrating that

this integrated approach has good flexibility in dealing with rapidly changing CRs [37]. ANP and fuzzy mathematics were integrated into QFD to develop new equipment for squeezing polyethylene pipes in a real-world scenario, demonstrating that multi-criteria decision-making and fuzzy algorithms can enhance the precision of the QFD process [14].

In summary, QFD can translate customer needs into engineering features in product development, thus potentially making it a valuable tool for SMEs to integrate CR information when screening concept sketches for PI. However, the traditional HoQ demands extensive data on CRs and ECs and is time-consuming to process, which hampers accurate decision-making with fewer experts. Moreover, the single-point values in the relationship matrix assessment fail to accurately capture the evaluators' subjective and fuzzy information.

2.4. Grey Number

SMEs face issues with incomplete and inaccurate information affecting decision-making due to limited human and resource inputs in NPD. In 1989, Professor Deng introduced the Grey System Theory, which is used to address problems in systems with small amounts of data and partially unknown information [38]. Grey Relational Analysis (GRA) was employed to study the deviations in supplier rating results due to the uncertainty and limited availability of information during the supplier selection process. By comparing GRA with the AHP and the ANP, they validated the effectiveness of GRA in addressing this issue [39]. An improved method combining GRA and AHP was proposed for effectively selecting the optimal design scheme for the Green Jack-up Drilling Platform (GJDP), and the results verified the feasibility and effectiveness of using the combined GRA and AHP methods [40]. A hybrid method combining GRA and MCDM was proposed to provide rational and effective support for the selection decision of interior decoration materials, verifying the effectiveness of the integrated application of GRA and AHP methods [41]. An improved QFD method was proposed to introduce interval grey numbers, assisting product developers in extracting valuable information from limited and imprecise market research or expert opinions and accurately identifying key engineering development recommendations [17].

In this study, grey numbers are introduced into KE and QFD respectively, used for calculating the AHP of the experts' evaluation of Kansei weights and for assessing the mapping relationship between Kansei CRs and ECs, with the aim of improving the accuracy of data processing for small samples and uncertain information.

3. Materials and Methods

This paper aims to introduce an innovative approach that helps SMEs maintain the PI elements in product form design while also satisfying the emotional needs of users. Furthermore, it seeks to achieve precise decision-making outcomes even under conditions of limited resources, such as small samples and scarce information. This method mainly consists of four phases, as shown in Figure 2. In the first phase, representative products of a certain brand are collected online and offline, the typical PI form elements are analyzed and defined, and then conceptual sketch alternatives are generated through training and iteration using SG and an AI-powered tool (Midjourney). In the second phase, Kansei words are extracted from online reviews, Kansei factors are then identified through FA, the hierarchical structure of Kansei requirements is constructed by employing the G-AHP, and the grey weights of finial consumer requirements (FCRs) are calculated. In the third phase, generated alternatives in the first phase are screened, and the PIECs of key forms are deconstructed using MA. Meanwhile, the PIECs, along with the FCRs, are input into the QFD model. The mapping relationship between them is scored by experts using grey interval numbers, and the ranking results for the PIECs of each key form are calculated. In the fourth phase, the optimal PIECS sketch alternatives are chosen and utilized for the detailed design and rendering of the final scheme, and the results are verified.

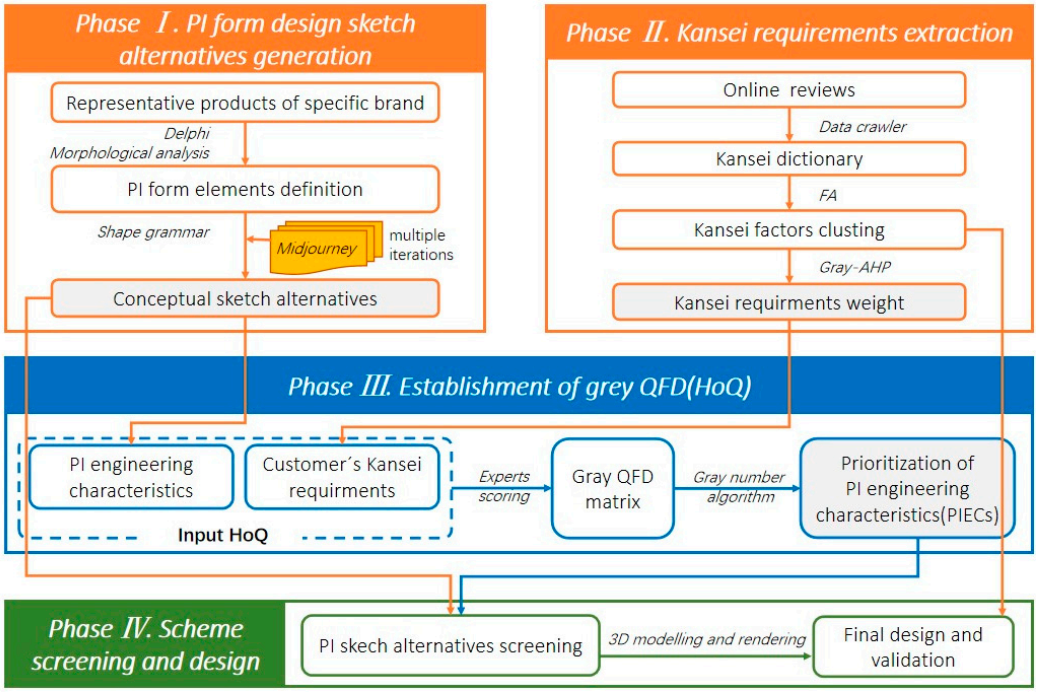


Figure 2. The research framework for product form design of PI and KE.

3.1. PI Form Design Sketch Alternative Generation

The generation of conceptual sketches is an extremely challenging task in product form design, which demands a high level of time consumption and capability from the design team. Additionally, the retention of PI elements within the sketches further complicates the task, which is unfriendly to SMEs. In this phase, a method combining SG with Midjourney is proposed, which is capable of generating form designs that retain PI characteristics and, based on AIGC technology, rapidly produces a multitude of conceptual sketch alternatives. The steps are as follows:

Step 1: Defining PI characteristics. Through online and offline surveys, collect typical products of a certain brand and obtain a product set in the form of images. Inviting design experts to form a team, using the Delphi and MA to screen the product set, obtain significant PI elements, and define them to get element descriptions and schematic diagrams.

Step 2: SG + Midjourney. According to Stiny, G., the shape grammar G = (S, L, R, I) consists of four components: (1) S is a finite set of shapes, (2) L is a finite set of labels, (3) R is a finite set of shape rules of the form $\alpha \rightarrow \beta$, where α is a labeled shape in (S, R)+ and β is a labeled shape in (S, R)*, and (4) I is a label shape in (S, R)+ called the initial shape [42].

The key technology of SG is transformation rules, which include generative rules for creating something from nothing based on the initial form, and modification rules for performing operations such as stretching, scaling, translation, rotation, deformation, addition and subtraction, replacement, and duplication [9,19]. In practice, the complexity of parameterized encoding in SG, along with the need for manual constraint editing and high computational requirements, means that the method's application is still limited and under exploration.

Midjourney is an AI creative platform with the advantage that users can input prompts to generate text-to-image and image-to-image processes. In the fields of design and artistic creation, this technology has gradually been adopted, becoming a potential creative tool for designers and artists [43]. By integrating text-to-image and image-to-image technologies, an innovative color block annotation method is used for pre-processing the image sources of PI. Midjourney is then utilized for image recognition training, combined with the 8 transformation rules from SG, and input into the Midjourney Model Version 6.0 as prompts, which is the current default model. Through the modification of the prompts

(transformation rules and image source links) and multiple iterations, a large number of conceptual sketch alternatives are ultimately obtained.

3.2. Kansei CR Extraction

In this phase, the focus is on the obtaining and weighting of users' emotional needs. Meeting the emotional needs of users is an important factor in determining whether a new product will be favored by the market. The methods of KE and G-AHP are used for the acquisition and calculation of users' emotional needs.

Step 3: Mining data. Use Octoparse v8.7.0 to crawl user reviews of products on e-commerce platforms, since the Kansei words for emotional needs are mostly adjectives, the high-frequency adjectives with positive semantic explanations are screened out to obtain a Kansei dictionary.

Step 4: Clustering Kansei words. Select products with distinct characteristics available on the market and use images as samples to incorporate representative emotional words from the Kansei dictionary for questionnaire design. Invite more than 10 experts with over 5 years of experience in E-moped design and product design to form an expert team to score the questionnaire and use FA to process the data, clustering to identify important Kansei factors. Data analysis is conducted using the online application software Statistical Product and Service Software Automatically (SPSSAU Version 23.0).

Step 5: G-AHP calculating Kansei weights. Combine grey numbers and AHP to calculate the weights of Kansei factors. Firstly, use grey numbers to replace the single-point value linguistic scales that represent semantic differences in traditional AHP methods for expert questionnaire design, as shown in Table 1. Secondly, construct a grey pairwise comparison matrix: Equation (1) represents the judgment of expert e when making a grey pairwise comparison between the ith and jth indicators. By having expert e compare all pairs of indicators, a grey pairwise comparison matrix is formed, as given in Equation (2). Thirdly, calculate the geometric mean of all the grey pairwise comparison matrices provided by the experts, as given in Equation (3). The difference in G-AHP is to calculate the geometric means for both the upper and lower bounds of the interval grey numbers separately. After the combination of the experts' pairwise comparison, the main pairwise comparison matrix D is found, as given in Equation (4). Fourthly, normalize the grey pairwise comparison matrix, as given in Equations (5) and (6), by calculating the upper and lower bounds of the grey interval matrix separately to obtain the normalized grey pairwise comparison matrix D*, as given in Equation (7). Fifthly, calculate the final grey weights of the Kansei CRs, as given in Equation (8), where $\otimes FCR_i$ represents the final grey weight of requirement i. The obtained grey weight results are used to rank the FCRs.

$$\bigotimes X_{ij}^{e} = \left[\underline{X}_{ij}^{e}, \overline{X}_{ij}^{e}\right] \tag{1}$$

$$D^{e} = \begin{bmatrix} \bigotimes X_{11}^{e} & \cdots & \bigotimes X_{1n}^{e} \\ \vdots & \ddots & \vdots \\ \bigotimes X_{n1}^{e} & \cdots & \bigotimes X_{nn}^{e} \end{bmatrix} \tag{2}$$

$$\bigotimes X_{ij} = \sqrt[D]{\prod_{d=1}^{D} \bigotimes X_{ij}^{d}} \tag{3}$$

$$D = \begin{bmatrix} \bigotimes X_{11} & \cdots & \bigotimes X_{1n} \\ \vdots & \ddots & \vdots \\ \bigotimes X_{n1} & \cdots & \bigotimes X_{nn} \end{bmatrix} \tag{4}$$

$$\underline{X_{ij}^{*}} = \left[2\underline{X}_{ij} / \left(\sum_{i=1}^{n} \underline{X}_{ij} + \sum_{i=1}^{n} \overline{X}_{ij}\right)\right] \tag{5}$$

$$\overline{X_{ij}^{*}} = \left[2\overline{X}_{ij} / \left(\sum_{i=1}^{n} \underline{X}_{ij} + \sum_{i=1}^{n} \overline{X}_{ij}\right)\right] \tag{6}$$

$$D^* = \begin{bmatrix} \otimes X_{11}^* & \cdots & \otimes X_{1n}^* \\ \vdots & \ddots & \vdots \\ \otimes X_{n1}^* & \cdots & \otimes X_{nn}^* \end{bmatrix} \quad (7)$$

$$\otimes FCR_i = \frac{1}{n}\sum_{i=1}^{n}\left[\underline{X}_{ij}^*, \overline{X}_{ij}^*\right] \quad (8)$$

Table 1. Linguistic scales and grey numbers used for the pairwise comparisons of the G-AHP.

Importance Value	Linguistic Scale	Grey Number
1	Equally Important	[1, 2]
3	Weakly Important	[2, 4]
5	Important	[4, 6]
7	Strongly Important	[6, 8]
9	Absolutely Important	[8, 10]

3.3. G-QFD

In this phase, construct a QFD framework to obtain a FCR and PIEC correlation judgment matrix, and invite experts to score the correlation using interval grey numbers, resulting in the HoQ of ECs for each PI element. The correlation results between FCRs and PIECs are calculated to determine the priority ranking of PIECs.

Step 6: Constructing G-QFD judgment matrixes. Inpu the FCR results obtained from step 5 into the left wall of the HoQ, while inviting experts from the fields of industrial design, brand strategy, engineering manufacturing, and so on to form an expert team. Then, using KJ and MA, the important PI form design elements are deconstructed to obtain multiple PIECs, which are then input into the roof of each HoQ, respectively. The operation process is shown in Figure 3.

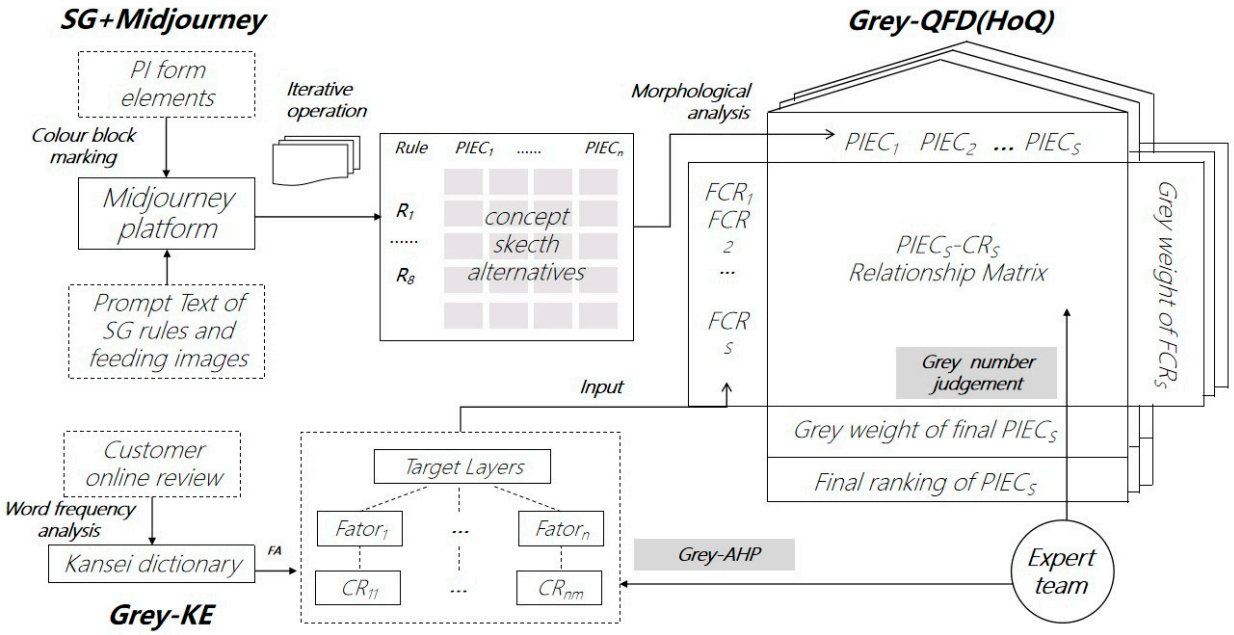


Figure 3. The process of inputting PIECs and FCRs into QFDs.

Step 7: Calculating the final weight of PIECs. By implementing the QFD process, the expert team is further invited to assess the level of correlation between FCRs and PIECs using grey numbers. The final importance calculation of PIECs is given in Equation (9), where $\otimes PIEC_i$ represents the grey weight of the final PIECs and $\otimes R_{ij}$ represents the level of correlation between FCRs and PIECs, evaluated using an interval grey number [0, 1]:

$$\begin{aligned}\otimes PIEC_i &= \sum_{i=1}^{n}\left(\otimes FCR_i \cdot \otimes R_{ij}\right), i, j = 1, 2, \ldots, n \\ &= \left[\min\left\{\underline{FCR_1}\cdot\underline{R_{1j}}, \underline{FCR_1}\cdot\overline{R_{1j}}, \overline{FCR_1}\cdot\underline{R_{1j}}, \overline{FCR_1}\cdot\overline{R_{1j}}\right\}, \max\left\{\underline{FCR_1}\cdot\underline{R_{1j}}, \underline{FCR_1}\cdot\overline{R_{1j}}, \overline{FCR_1}\cdot\underline{R_{1j}}, \overline{FCR_1}\cdot\overline{R_{1j}}\right\}\right] \\ &+ \left[\min\left\{\underline{FCR_2}\cdot\underline{R_{2j}}, \underline{FCR_2}\cdot\overline{R_{2j}}, \overline{FCR_2}\cdot\underline{R_{2j}}, \overline{FCR_2}\cdot\overline{R_{2j}}\right\}, \max\left\{\underline{FCR_2}\cdot\underline{R_{2j}}, \underline{FCR_2}\cdot\overline{R_{2j}}, \overline{FCR_2}\cdot\underline{R_{2j}}, \overline{FCR_2}\cdot\overline{R_{2j}}\right\}\right] \\ &\ldots + \left[\min\left\{\underline{FCR_n}\cdot\underline{R_{nj}}, \underline{FCR_n}\cdot\overline{R_{nj}}, \overline{FCR_n}\cdot\underline{R_{nj}}, \overline{FCR_n}\cdot\overline{R_{nj}}\right\}, \max\left\{\underline{FCR_n}\cdot\underline{R_{nj}}, \underline{FCR_n}\cdot\overline{R_{nj}}, \overline{FCR_n}\cdot\underline{R_{nj}}, \overline{FCR_n}\cdot\overline{R_{nj}}\right\}\right] \\ &= \left[\underline{PIEC_i}, \overline{PIEC_i}\right]\end{aligned} \tag{9}$$

Step 8: Determine the final ranking order of PIECs. Since the importance level of CRs and ECs is represented as an interval grey number, assume two interval grey numbers: $\otimes GN_1 = \left[\underline{GN_1}, \overline{GN_1}\right]$ and $\otimes GN_2 = \left[\underline{GN_2}, \overline{GN_2}\right]$, $\otimes GN_1 > 0$, $\otimes GN_2 > 0$. There are six possible conditions between $\otimes GN_1$, and $\otimes GN_2$, as shown in Figure 4.

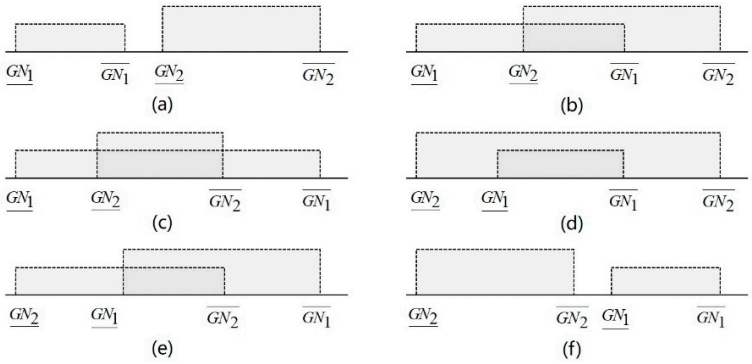


Figure 4. Six possible scenarios (**a**–**f**) of $\otimes GN_1$ and $\otimes GN_2$.

Scholars use the geometric area comparison method in traditional vector spaces to rank grey numbers by comparing the areas in the vector space [44]. In Figure 4, the results for (a) and (f) can be clearly derived; hence, the vector space is redrawn, taking (b–e) as an example, as shown in Figure 5.

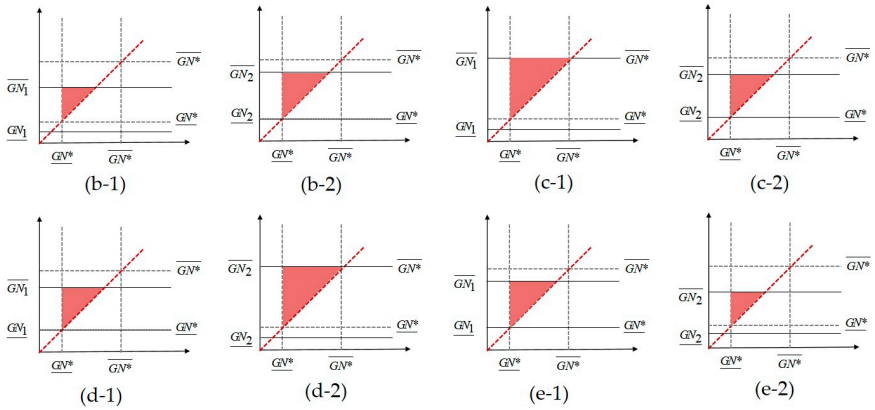


Figure 5. Areas of 8 vector spaces.

Firstly, derive an ideal grey number $\otimes GN^*$, as given in Equation (10):

$$\otimes GN^* = \left[max\left(\underline{GN_1}, \underline{GN_2}\right), max\left(\overline{GN_1}, \overline{GN_2}\right)\right] \tag{10}$$

Then, calculate the area $S\left(\otimes GN_1, \otimes GN^*\right)$, among the $\otimes GN_1, \otimes GN^*$, and 45° line, and similarly calculate the area $S\left(\otimes GN_2, \otimes GN^*\right)$, as given in Equations (11) and (12):

$$S\left(\otimes GN_1, \otimes GN^*\right) = \frac{1}{2}\left(\overline{GN_1} - \underline{GN^*}\right)^2 \tag{11}$$

$$S\left(\bigotimes GN_2, \bigotimes GN^*\right) = \frac{1}{2}\left(\overline{GN_2} - \underline{GN^*}\right)^2 \quad (12)$$

The priority of $\bigotimes GN_1$ and $\bigotimes GN_2$ is defined as follows:

If (a) $S(\bigotimes GN_1, \bigotimes GN^*) < S(\bigotimes GN_2, \bigotimes GN^*)$, then $\bigotimes GN_1 < \bigotimes GN_2$;
or, if (b) $S(\bigotimes GN_1, \bigotimes GN^*) = S(\bigotimes GN_2, \bigotimes GN^*)$, then $\bigotimes GN_1 = \bigotimes GN_2$;
or, if (c) $S(\bigotimes GN_1, \bigotimes GN^*) > S(\bigotimes GN_2, \bigotimes GN^*)$, then $\bigotimes GN_1 > \bigotimes GN_2$.

4. Case Validation and Results

Previous studies have indicated that the design of transportation products puts more emphasis on PI. Whether it is cars or motorcycles, brands such as BMW, Buick, and Harley-Davidson have all received attention in the field of academic research [21,22]. This section selects a Chinese brand of female E-mopeds as an example to verify the feasibility of the proposed method. At present, the brand's products have initially established the PI design. However, since early E-moped users were predominantly male, the product form designs were mainly led by male aesthetics. Currently, the female E-mopeds on sale only meet the aesthetic demands through modifications such as color, painting, structural lightweighting, and accessories, rather than starting from the initial CRs, as shown in Figure 6.



Figure 6. Three types of modifications for female E-mopeds.

Since females focus more on emotional needs when making purchasing decisions, integrating feminine emotional demands into product form design has become a key element of the design process [45]. It is a significant challenge for SMEs to design a product form that retains PI while meeting female CRs efficiently and accurately with limited resources available. To achieve this goal, the process is specifically divided into the following three phases:

The first phase: Generation of conceptual alternatives for PI form design

Step 1: Defining PI characteristics. The form of E-mopeds has two different riding postures: sitting-style and straddle-style. The brand's mainstream sitting-style product was selected by this study. Previous studies on structurally similar vehicles such as motorcycles or mopeds had primarily used the side view as the main perspective for research [46]. Therefore, we chose the side view as the research perspective and collected 25 product images, online and offline. An expert team of five experts, including E-moped specialists, brand strategists, and industrial designers, utilized the KJ method to select a representative sample (model NQi). Eight form design features were extracted and coded A–H, as shown in Figure 7.

Using Photoshop, the images of 25 products were refined to eliminate color and pattern distractions. The MA and Delphi methods were used by experts to identify 3 main PI form elements that influence the product design: (1) Front Cover: an arrow-shaped front cover that extends to the sides (Figure 8a). (2) Seat: the seat contour is more rectangular, avoiding free curves (Figure 8b). (3) Side Cover: the side bottom cover and side body cover are polygonal in shape, divided into separate and integrated styles (Figure 8c,d).

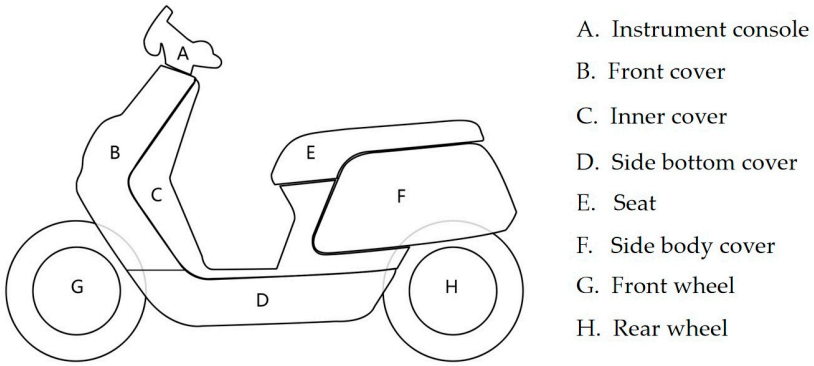


Figure 7. The form design features of the E-moped.

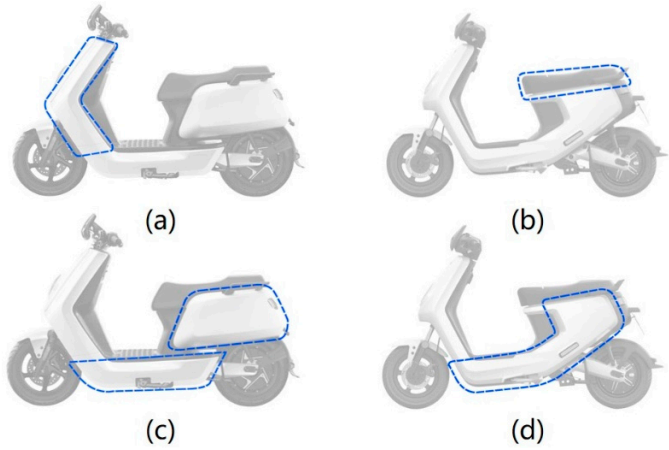


Figure 8. Three main PI form elements.

Step 2: Generating conceptual sketch alternatives. Midjourney cannot effectively recognize the PI form elements of images. First, the PI area of the product image was manually marked with color blocks for pre-processing and then input into Midjourney for training. Taking one side body cover as an example, the pre-trained image obtained after manual processing was shown in Figure 9, where the red area indicated the PI form area for AI recognition training.



Figure 9. Pre-training image of the side body cover.

Second, the pre-trained images were input into Midjourney to obtain image links, and then one of the 8 deductive rule prompts was input for image generation. After several iterations and selections, the concept sketches were obtained. Taking an iteration-generated interface as an example, the U button and V button were used to select and fine-tune the images for the next round of generation. At the same time, the prompts can also be modified through the bottom dialogue box to obtain other alternatives of PI, as shown in Figure 10.

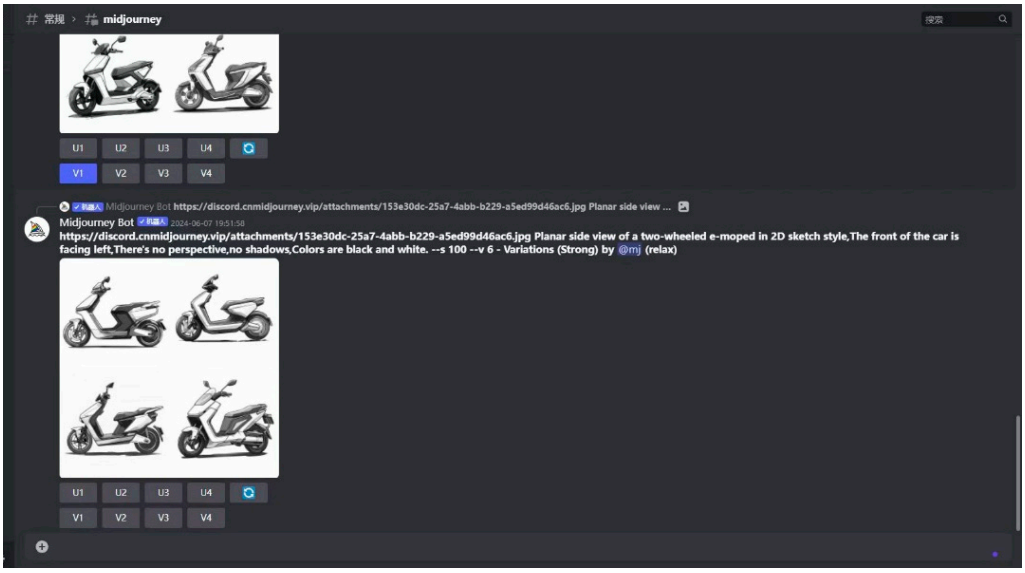


Figure 10. The interface of Midjourney for sketch generation.

Last, this study obtained 63 initial concept sketches, from which the expert group selected and removed those that clearly did not conform to the PI and those that had similar designs, resulting in 48 concept sketch alternatives, as shown in Figure 11.

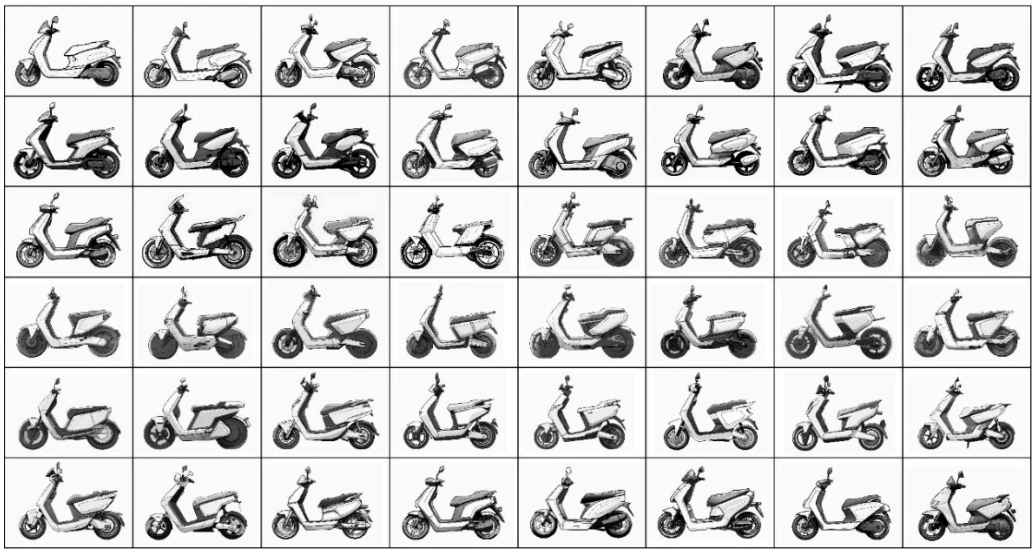


Figure 11. The side view of screened sketch alternatives.

The second phase: Extraction of E-moped Kansei CRs

Step 3: Constructing the Kansei dictionary. The Chinese online shopping platform JD.com and the American platform Amazon.com are both large-scale platforms with a vast amount of reviews [5]. To ensure the diversity of data collection, the Octopus tool was used to crawl data from the two platforms, selecting review data from mainstream brands such as Yade, Aima, Niu, and Luyuan under the female E-moped category [47], and a total of 6181 reviews were collected. After filtering out duplicates, meaningless content, and expressions that only conveyed good or bad sentiments, a Kansei dictionary consisting of 72 words with a frequency higher than 50 was constructed, as shown in Table 2.

Table 2. Seventy-two Kansei words and their frequency.

Kansei Word	Frequency	Kansei Word	Frequency	Kansei Word	Frequency	Kansei Word	Frequency	Kansei Word	Frequency
good	4786	small	351	enthusiastic	187	enough	100	free	65
electric	1137	long	342	effective	177	light	98	useful	63
appearance	1077	performance	321	trustworthy	169	smooth	97	careful	61
fast	1023	pretty	281	particularly	167	attractive	96	cheap	60
beautiful	851	excellent	280	patient	151	endurance	95	sturdy	60
quality	816	strong	270	sufficient	135	positive	91	pop	57
high	687	stable	236	intelligent	127	right	80	happy	56
speed	647	suitable	235	simple	122	professional	79	expected	54
satisfied	645	shock	231	durable	120	decent	77	amazing	54
convenient	528	big	216	perfect	112	cool	75	national	53
easy	518	lightweight	213	helped	109	friendly	74	forward	53
experience	513	new	211	compact	109	fashionable	72	safe	52
powerful	498	affordable	207	exquisite	108	satisfying	72		
comfortable	444	large	188	cute	105	practical	69		
worth	423	awesome	187	handsome	103	stylish	68		

Step 4: Clustering Kansei factors. Cluster the Kansei words using FA to identify the user's concerns. A focus group of 15 E-moped and product design experts with over 5 years of experience screened female E-moped search results on JD and Amazon, and then applied the KJ method to refine the Kansei dictionary to 10 Kansei words, as shown in Table 3.

Table 3. Ten Kansei words obtained by the KJ method.

beautiful	convenient	comfortable	stable	lightweight
durable	compact	cute	fashionable	safe

Since this study explores the impact of emotional needs on form design, Photoshop was used to desaturate product images and eliminate the interference of background and perspective differences in E-mopeds. Three selected products with significant differences are shown in Figure 12.



Figure 12. Three selected products.

The obtained 10 Kansei words and 3 typical samples were used to create a conformity questionnaire, which was then distributed to female consumers for scoring. The questionnaires were designed on a 5-point Likert scale with a score of 1 for not conforming, 2 for less conforming, 3 for average, 4 for more conforming, and 5 for conforming. The questionnaire collected a total of 60 subjects, of which 30 were female students majoring in product design, and the other 30 were middle-aged and young women from society.

Data analysis was conducted using the SPSSAU software, and the Cronbach's α coefficient was 0.934, indicating the high reliability of the questionnaire data. The Kaisere-Meyere-Olkin (KMO) measure and Bartlett's test of sphericity were conducted, resulting in a KMO value of 0.754, indicating that FA can be used to process the data. The Bartlett's test

of sphericity value was $p = 0.000$, indicating the data was spherically distributed, which means that the variables were not independently distributed and were suitable for FA.

FA was conducted on the questionnaire data for the clustering of Kansei words. When three clustering factors were selected, the cumulative variance interpretation rate was 72.105%; thus, the factors were extracted into 3. The maximum variance method was applied to the factors to obtain the rotated factor matrix, as shown in Table 4. Absolute values of factor loading coefficients below 0.5 were omitted from the display in the table for clarity. Factor 1 consists of 4 Kansei words: "lightweight", "compact", "cute", and "fashionable", which name the aesthetic factor. Factor 2 consists of "beautiful", "convenient", and "comfortable", which name the affinity factor. Factor 3 consists of "stable", "durable", and "safe", which name the trust factor.

Table 4. Rotated factor matrix.

Ingredient	Factor 1	Factor 2	Factor 3	Kansei Factor
lightweight	0.771			aesthetic factor
compact	0.773			
cute	0.767			
fashionable	0.691			
beautiful		0.742		affinity factor
convenient		0.888		
comfortable		0.736		
stable			0.742	trust factor
durable			0.827	
safe			0.763	

Step 5: Calculating Kansei CRs weights. Grey theory was integrated into the AHP to construct a 3-layer Kansei CR model, with females' Kansei CRs for E-mopeds as the target layer, 3 Kansei factors as the criterion layer, and 10 Kansei words as the sub-criterion layer. Twelve experts were invited to make judgments on perceptual demands, including 2 design directors with more than 8 years of industrial design experience, 3 industrial design professors, and 7 master's degree graduate students majoring in industrial design, with 50% being female. Experts conducted a questionnaire interview in the form of focus groups to obtain the two-by-two pairwise comparison matrix for each expert's judgment on the criteria and sub-criteria.

Taking expert 1 as an example, judgment matrix A_1 was obtained using Equations (1) and (2), and interval grey numbers were used to value the judgment matrix.

$$A_1 = \begin{pmatrix} [1,1] & [2,4] & [4,6] \\ [1/4,1/2] & [1,1] & [2,4] \\ [1/6,1/4] & [1/4,1/2] & [1,1] \end{pmatrix}$$

The judgment results from the 12 experts were calculated using Equations (3) and (4) to obtain the main pairwise comparison matrix and weights, as shown in Table 5.

Table 5. Main comparison matrix and grey weights of Kansei factors.

Kansei Factor	Code	C1	C2	C3	Grey Weight	Crisp Weight
aesthetic factor	C1	[1.000, 1.000]	[1.024, 1.815]	[0.958, 1.407]	[0.324, 0.447]	0.386
affinity factor	C2	[0.551, 0.976]	[1.000, 1.000]	[0.794, 1.367]	[0.248, 0.361]	0.304
trust factor	C3	[0.711, 1.043]	[0.731, 1.260]	[1.000, 1.000]	[0.263, 0.357]	0.310

According to Equations (5)–(8), the judgment results of the sub-criterion layer were calculated in sequence, and the initial grey weight results of each of the Kansei words are shown in Tables 6–8.

Table 6. Initial grey weight of the aesthetic factors' sub-criteria.

Kansei Word	Code	C11	C12	C13	C14	Grey Weight	Crisp Weight
beautiful	C11	[1.000, 1.000]	[0.338, 0.569]	[0.371, 0.624]	[0.176, 0.275]	[0.083, 0.120]	0.101
compact	C12	[1.757, 2.954]	[1.000, 1.000]	[0.507, 0.821]	[0.403, 0.700]	[0.167, 0.247]	0.207
cute	C13	[1.603, 2.696]	[1.218, 1.972]	[1.000, 1.000]	[0.386, 0.645]	[0.201, 0.293]	0.247
fashionable	C14	[3.634, 5.682]	[1.513, 2.484]	[1.550, 2.592]	[1.000, 1.000]	[0.365, 0.525]	0.445

Table 7. Initial grey weight of the affinity factor's sub-criteria.

Kansei Word	Code	C21	C22	C23	Grey Weight	Crisp Weight
lightweight	C21	[1.000, 1.000]	[0.661, 1.189]	[0.380, 0.684]	[0.194, 0.287]	0.240
convenient	C22	[0.841, 1.513]	[1.000, 1.000]	[0.342, 0.630]	[0.203, 0.303]	0.253
comfortable	C23	[1.463, 2.632]	[1.587, 2.926]	[1.000, 1.000]	[0.407, 0.606]	0.507

Table 8. Initial grey weight of the trust factor's sub-criteria.

Kansei Word	Code	C31	C32	C33	Grey Weight	Crisp Weight
stable	C31	[1.000, 1.000]	[0.519, 0.891]	[0.157, 0.222]	[0.107, 0.142]	0.125
durable	C32	[1.122, 1.925]	[1.000, 1.000]	[0.187, 0.294]	[0.145, 0.204]	0.175
safe	C33	[4.427, 6.504]	[3.397, 5.334]	[1.000, 1.000]	[0.604, 0.797]	0.701

Normalization calculations were performed to obtain the final weights for each Kansei word, with FCR_1 to FCR_{10} encoded for the 10 Kansei CRs, as shown in Table 9.

Table 9. Finial grey weights of CRs.

Code	Grey Weight	Crisp Weight	Code	Grey Weight	Crisp Weight
FCR_1	[0.0268, 0.0536]	0.039	FCR_6	[0.0503, 0.1093]	0.077
FCR_2	[0.0541, 0.1106]	0.080	FCR_7	[0.1011, 0.2185]	0.154
FCR_3	[0.0642, 0.1314]	0.095	FCR_8	[0.0281, 0.0508]	0.039
FCR_4	[0.1185, 0.2352]	0.172	FCR_9	[0.0383, 0.0727]	0.054
FCR_5	[0.0482, 0.1033]	0.073	FCR_{10}	[0.1591, 0.2845]	0.217

The third phase: Establishment of G-QFD

Step 6: Establishing HoQs of FCRs and PIECs. The team of experts, consisting of 10 E-mopeds and previously invited industrial design professionals, deconstructed the 48 conceptual sketch alternatives using MA. Combined with the 8 PI form design features obtained earlier, they ultimately identified 3 categories (Table 10):

1. Side cover PIECs: 14 different forms of side button covers and side body covers.
2. Front cover PIECs: 8 different forms of front covers.
3. Seat PIECs: 5 different forms of seats.

Step 7: Calculating the initial correlation level of PIECs within each category. The relationship judgment between FCRs and PIECs by the 12 experts from step 5 using interval numbers [0, 1] was scored to obtain 3 HoQ relationship judgment matrices. Their calculated grey weights are shown in Tables 11–13.

Step 8: Calculating the final grey weights for PIECs within each category. According to Equations (10)–(12), the final grey weights and priority rankings of each PIEC in Tables 11–13 were calculated, as shown in Tables 13–15. The optimal engineering characteristic in the Side cover PIECs was the 7th, with the grey weight value of [0.307, 0.752]. In the Front cover PIECs, the optimal engineering characteristic is the 3rd, with the grey weight value of [0.303, 0.769]. In the Seat PIECs, the optimal engineering characteristic is the 4th one, with the grey weight value of [0.321, 0.776], as shown in Tables 14–16.

Table 10. PIECs of the three PI form features.

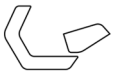
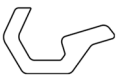
PIECs	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Side cover								
	9	**10**	**11**	**12**	**13**	**14**		
								
PIECs	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Front cover								
PIECs	**1**	**2**	**3**	**4**	**5**			
Seat								

Table 11. Side cover grey relationship matrices between the FCRs and the PIECs.

Side Cover								
	$PIEC_1$	**$PIEC_2$**	**$PIEC_3$**	**$PIEC_4$**	**$PIEC_5$**	**$PIEC_6$**	**$PIEC_7$**	**$PIEC_8$**
FCR_1	[0.04, 0.10]	[0.50, 0.58]	[0.46, 0.56]	[0.52, 0.61]	[0.36, 0.46]	[0.16, 0.23]	[0.34, 0.44]	[0.18, 0.31]
FCR_2	[0.02, 0.06]	[0.18, 0.27]	[0.21, 0.30]	[0.65, 0.74]	[0.24, 0.34]	[0.06, 0.10]	[0.34, 0.42]	[0.08, 0.10]
FCR_3	[0.04, 0.06]	[0.23, 0.32]	[0.20, 0.26]	[0.14, 0.20]	[0.15, 0.23]	[0.42, 0.52]	[0.64, 0.74]	[0.06, 0.11]
FCR_4	[0.30, 0.40]	[0.26, 0.36]	[0.25, 0.35]	[0.22, 0.32]	[0.69, 0.78]	[0.67, 0.76]	[0.78, 0.87]	[0.12, 0.14]
FCR_5	[0.36, 0.46]	[0.54, 0.64]	[0.41, 0.54]	[0.56, 0.66]	[0.16, 0.24]	[0.14, 0.16]	[0.66, 0.76]	[0.22, 0.30]
FCR_6	[0.16, 0.20]	[0.36, 0.52]	[0.24, 0.34]	[0.34, 0.40]	[0.18, 0.28]	[0.34, 0.42]	[0.42, 0.52]	[0.70, 0.80]
FCR_7	[0.66, 0.76]	[0.28, 0.38]	[0.34, 0.42]	[0.32, 0.42]	[0.18, 0.28]	[0.46, 0.55]	[0.48, 0.58]	[0.42, 0.52]
FCR_8	[0.72, 0.82]	[0.15, 0.17]	[0.16, 0.18]	[0.27, 0.37]	[0.04, 0.06]	[0.46, 0.56]	[0.16, 0.24]	[0.58, 0.68]
FCR_9	[0.74, 0.84]	[0.22, 0.32]	[0.23, 0.33]	[0.36, 0.44]	[0.06, 0.08]	[0.50, 0.60]	[0.14, 0.20]	[0.58, 0.67]
FCR_{10}	[0.76, 0.85]	[0.24, 0.38]	[0.24, 0.34]	[0.40, 0.50]	[0.06, 0.08]	[0.45, 0.57]	[0.22, 0.32]	[0.34, 0.44]
	$PIEC_9$	**$PIEC_{10}$**	**$PIEC_{11}$**	**$PIEC_{12}$**	**$PIEC_{13}$**	**$PIEC_{14}$**	**Grey weight**	
FCR_1	[0.48, 0.58]	[0.53, 0.63]	[0.06, 0.10]	[0.04, 0.06]	[0.67, 0.76]	[0.59, 0.68]	[0.0268, 0.0536]	
FCR_2	[0.26, 0.36]	[0.28, 0.36]	[0.00, 0.00]	[0.06, 0.08]	[0.52, 0.62]	[0.44, 0.56]	[0.0541, 0.1106]	
FCR_3	[0.12, 0.16]	[0.10, 0.18]	[0.06, 0.08]	[0.04, 0.06]	[0.18, 0.28]	[0.11, 0.20]	[0.0642, 0.1314]	
FCR_4	[0.36, 0.42]	[0.22, 0.32]	[0.08, 0.14]	[0.40, 0.54]	[0.28, 0.40]	[0.46, 0.56]	[0.1185, 0.2352]	
FCR_5	[0.34, 0.43]	[0.14, 0.18]	[0.06, 0.10]	[0.12, 0.16]	[0.38, 0.52]	[0.28, 0.42]	[0.0482, 0.1033]	
FCR_6	[0.58, 0.68]	[0.64, 0.73]	[0.16, 0.20]	[0.08, 0.14]	[0.40, 0.52]	[0.28, 0.37]	[0.0503, 0.1093]	
FCR_7	[0.14, 0.20]	[0.20, 0.28]	[0.74, 0.83]	[0.60, 0.70]	[0.26, 0.32]	[0.24, 0.32]	[0.1011, 0.2185]	
FCR_8	[0.10, 0.16]	[0.08, 0.12]	[0.66, 0.76]	[0.74, 0.83]	[0.28, 0.40]	[0.32, 0.42]	[0.0281, 0.0508]	
FCR_9	[0.30, 0.40]	[0.12, 0.16]	[0.52, 0.64]	[0.62, 0.74]	[0.28, 0.36]	[0.30, 0.38]	[0.0383, 0.0727]	
FCR_{10}	[0.04, 0.06]	[0.14, 0.20]	[0.76, 0.85]	[0.71, 0.88]	[0.24, 0.32]	[0.35, 0.47]	[0.1591, 0.2845]	

Table 12. Front cover grey relationship matrices between the FCRs and the PIECs.

	Front Cover								
	$PIEC_1$	**$PIEC_2$**	**$PIEC_3$**	**$PIEC_4$**	**$PIEC_5$**	**$PIEC_6$**	**$PIEC_7$**	**$PIEC_8$**	**Grey Weight**
FCR_1	[0.68, 0.80]	[0.28, 0.40]	[0.58, 0.70]	[0.24, 0.36]	[0.22, 0.36]	[0.28, 0.38]	[0.06, 0.14]	[0.02, 0.04]	[0.0268, 0.0536]
FCR_2	[0.58, 0.72]	[0.32, 0.44]	[0.52, 0.64]	[0.16, 0.26]	[0.12, 0.20]	[0.28, 0.40]	[0.06, 0.14]	[0.04, 0.08]	[0.0541, 0.1106]
FCR_3	[0.40, 0.52]	[0.46, 0.59]	[0.63, 0.76]	[0.25, 0.37]	[0.12, 0.24]	[0.28, 0.38]	[0.24, 0.32]	[0.22, 0.32]	[0.0642, 0.1314]
FCR_4	[0.42, 0.52]	[0.32, 0.44]	[0.56, 0.68]	[0.57, 0.69]	[0.33, 0.46]	[0.34, 0.44]	[0.36, 0.48]	[0.26, 0.36]	[0.1185, 0.2352]
FCR_5	[0.42, 0.54]	[0.42, 0.52]	[0.48, 0.58]	[0.52, 0.62]	[0.44, 0.54]	[0.42, 0.52]	[0.60, 0.71]	[0.29, 0.40]	[0.0482, 0.1033]
FCR_6	[0.63, 0.72]	[0.32, 0.40]	[0.50, 0.66]	[0.31, 0.41]	[0.20, 0.30]	[0.41, 0.51]	[0.12, 0.22]	[0.10, 0.16]	[0.0503, 0.1093]
FCR_7	[0.32, 0.42]	[0.35, 0.45]	[0.33, 0.41]	[0.38, 0.54]	[0.30, 0.42]	[0.50, 0.60]	[0.60, 0.74]	[0.42, 0.57]	[0.1011, 0.2185]
FCR_8	[0.28, 0.42]	[0.44, 0.56]	[0.34, 0.46]	[0.48, 0.58]	[0.27, 0.35]	[0.66, 0.78]	[0.63, 0.76]	[0.57, 0.70]	[0.0281, 0.0508]
FCR_9	[0.30, 0.40]	[0.34, 0.52]	[0.28, 0.42]	[0.50, 0.61]	[0.26, 0.34]	[0.61, 0.73]	[0.56, 0.66]	[0.48, 0.60]	[0.0383, 0.0727]
FCR_{10}	[0.24, 0.36]	[0.47, 0.60]	[0.32, 0.44]	[0.41, 0.58]	[0.33, 0.43]	[0.54, 0.66]	[0.58, 0.70]	[0.49, 0.61]	[0.1591, 0.2845]

Table 13. Seat grey relationship matrices between the FCRs and the PIECs.

	Seat					
	$PIEC_1$	**$PIEC_2$**	**$PIEC_3$**	**$PIEC_4$**	**$PIEC_5$**	**Grey Weight**
FCR_1	[0.38, 0.49]	[0.50, 0.60]	[0.60, 0.72]	[0.36, 0.48]	[0.20, 0.28]	[0.0268, 0.0536]
FCR_2	[0.28, 0.40]	[0.44, 0.56]	[0.52, 0.64]	[0.34, 0.44]	[0.12, 0.20]	[0.0541, 0.1106]
FCR_3	[0.34, 0.42]	[0.30, 0.38]	[0.31, 0.43]	[0.34, 0.42]	[0.34, 0.42]	[0.0642, 0.1314]
FCR_4	[0.56, 0.70]	[0.52, 0.62]	[0.56, 0.68]	[0.38, 0.50]	[0.22, 0.30]	[0.1185, 0.2352]
FCR_5	[0.42, 0.53]	[0.44, 0.55]	[0.56, 0.66]	[0.44, 0.54]	[0.27, 0.34]	[0.0482, 0.1033]
FCR_6	[0.36, 0.44]	[0.36, 0.45]	[0.47, 0.57]	[0.47, 0.60]	[0.38, 0.50]	[0.0503, 0.1093]
FCR_7	[0.48, 0.60]	[0.28, 0.44]	[0.28, 0.44]	[0.48, 0.58]	[0.56, 0.68]	[0.1011, 0.2185]
FCR_8	[0.41, 0.53]	[0.21, 0.31]	[0.22, 0.34]	[0.61, 0.71]	[0.71, 0.80]	[0.0281, 0.0508]
FCR_9	[0.44, 0.57]	[0.34, 0.46]	[0.38, 0.50]	[0.61, 0.70]	[0.70, 0.82]	[0.0383, 0.0727]
FCR_{10}	[0.44, 0.54]	[0.38, 0.50]	[0.39, 0.51]	[0.58, 0.68]	[0.75, 0.87]	[0.1591, 0.2845]

Table 14. Final grey important and ranking of side cover PIECs.

Side Cover	**$PIEC_1$**	**$PIEC_2$**	**$PIEC_3$**	**$PIEC_4$**	**$PIEC_5$**	**$PIEC_6$**	**$PIEC_7$**
Final important	[0.302, 0.694]	[0.192, 0.534]	[0.184, 0.494]	[0.246, 0.613]	[0.162, 0.424]	[0.288, 0.687]	[0.307, 0.752]
Ranking	2	9	11	5	12	3	1
Side cover	**$PIEC_8$**	**$PIEC_9$**	**$PIEC_{10}$**	**$PIEC_{11}$**	**$PIEC_{12}$**	**$PIEC_{13}$**	**$PIEC_{14}$**
Final important	[0.208, 0.516]	[0.158, 0.407]	[0.150, 0.407]	[0.260, 0.589]	[0.282, 0.678]	[0.212, 0.558]	[0.229, 0.593]
Ranking	10	13	14	7	4	8	6

Table 15. Final grey important and ranking of front cover PIECs.

Front Cover	**$PIEC_1$**	**$PIEC_2$**	**$PIEC_3$**	**$PIEC_4$**	**$PIEC_5$**	**$PIEC_6$**	**$PIEC_7$**	**$PIEC_8$**
Final important	[0.267, 0.692]	[0.264, 0.684]	[0.303, 0.769]	[0.276, 0.725]	[0.191, 0.526]	[0.300, 0.739]	[0.290, 0.723]	[0.221, 0.574]
Ranking	5	6	1	3	8	2	4	7

Table 16. Final grey important and ranking of seat PIECs.

Seat ECs	**PEC_1**	**PEC_2**	**PEC_3**	**PEC_4**	**PEC_5**
Final important	[0.299, 0.746]	[0.265, 0.683]	[0.292, 0.751]	[0.321, 0.776]	[0.315, 0.749]
Ranking	4	5	2	1	3

Based on the above results, the corresponding conceptual sketches were selected, as shown in the three red frame areas in Figure 13.

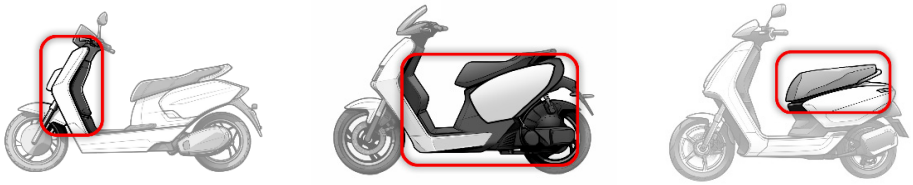


Figure 13. Three selected conceptual sketches.

The final design scheme for the female E-moped was obtained through 3D detailed design and final rendering from the selected conceptual sketches by senior designers using Rhino 6.0 and Keyshot 12, as shown in Figure 14.



Figure 14. Female E-moped design scheme.

The design scheme was utilized as a sample to conduct a questionnaire design for investigating and verifying the validity of the experimental results. The survey questionnaire was conducted using the Chinese online survey platform Wenjuanxing, where participants rated the questions via their mobile phones. Over a span of ten days, a total of 50 female subjects participated in the questionnaire survey, with ages ranging from 18 to 40. Among them, 23 were college students and 27 were other women in society. Before completing the questionnaire, they were required to view images of the design scheme and understand the emotional demand means of the 3 Kansei factors. A 7-point Likert scale was used for scoring, where 1 indicates very non-compliant, 2 indicates non-compliant, 3 indicates less compliant, 4 indicates fair, 5 indicates more compliant, 6 indicates compliant, and 7 indicates very compliant. An expert team also rated the final design's PI consistency, using the same scoring standards as before. The statistical results indicated that the affinity factor had the highest compliance score of 5.82, while the aesthetic factor and trust factor scored 5.62 and 5.60, respectively. The PI consistency compliance was relatively high at 5.7, as shown in Figure 15.

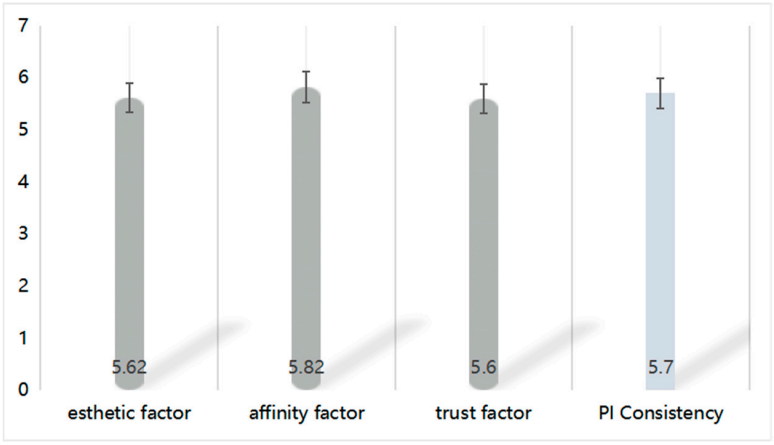


Figure 15. Compliant degree of Kansei factors and PI consistency.

5. Discussion

In this study, an AI-powered method is proposed, integrating SG and G-KE-QFD and utilizing Midjourney and web crawling, to address the challenge of integrating PI elements and users' emotional information in product form design for SMEs. Its effectiveness and feasibility have been verified by a female E-moped of a certain brand as an example.

Compared to existing research, the innovative methods proposed in this study have a uniqueness in three aspects of research: PI design, KE design, and AIGC design. Firstly, in PI design research, existing studies have effectively achieved consistency in PI features through the SG method. However, whether through the parameter-encoding method [21] or the label-encoding method [22], the complexity of a large number of rules makes execution very challenging. This current study, with AI-powered technology, has made the process more straightforward and user-friendly. Secondly, in KE design research, existing studies, although capable of effectively mining and mapping users' emotional needs through integrated methods, encounter problems such as high requirements for computer programming skills [5] or a lack of diversity in the generation of concept sketches [31]. This current study can efficiently achieve a diversity of sketches and precisely meet users' emotional needs. Finally, in AIGC design research, existing studies have explored the application value of AI in the conceptual design phase of architectural design [24], as well as AI's capability in product color-matching design [23], but they have not yet delved into the research of product form design. This current study has explored AI-powered product form design and verified its effectiveness. In summary, this study addresses the shortcomings mentioned in the above previous studies, as shown in Table 17.

Table 17. Summary of comparison with literature.

Classification	Description	Authors	Methods	Analysis
PI design	The application of SG in 2D design	Pugliese and Cagan [21]	SG/Parametric coding	Consistent branding, yet excessive rule parameters.
	The application of SG in 3D design	McCormack et al. [22]	SG/function lables	Consistent PI, yet excessive label rules.
	This current study		SG/Midjourney/ G-QFD/KE	Simpler rules, preserving PI design.
KE design	Computer programming in KE	Yang et al. [5]	DM/SVR/KE	Efficiently mining and filtering emotional needs, but requires a high level of computer programming skills.
	Intelligence algorithms in KE	Kang [31]	SVR/RST/QFD/KE	Higher fitting accuracy of mapping, but lacks the ability to quickly generate a large number of sketches.
	This current study		SG/Midjourney/ G-QFD/KE	Effectively acquires and processes KE requirements, precisely capturing needs and performing sketch screening.
AIAD design	AI application in the creative phase of architectural design	Hanafy [24]	AI/Text-to-Image/ NLP	Effective in architectural concept design, yet unexplored in product design.
	AI application in product color-matching design	Wu et al. [23]	AIGC/ChatGPT/ Midjourney	Effective in color design, but form design remains unexplored.
	This current study		SG/Midjourney/ G-QFD/KE	Effective application in product form design.

The result of this study has demonstrated the effectiveness of Midjourney in the creative stage of product PI form design, and the online reviews and analysis based on big data can help SMEs to compensate for their insufficient exploration of users' emotional needs. Integrating grey numbers into the AHP method and QFD framework effectively addresses the impact of subjectivity and fuzzy in decision-making and relationship judgment due to a small number of experts, which affects the precision of the results. Ultimately, this helps SMEs efficiently design schemes that incorporate PI elements and meet users' emotions.

In summary, this study makes the following contributions:

1. In this study, the integration of big data mining technology with AHP into the KE framework for obtaining and analyzing online review information ensures the com-

pleteness and objectivity of the extraction of CRs. Meanwhile, the introduction of grey numbers into AHP and QFD effectively addresses the issues of subjectivity and fuzziness in the case of a limited number of experts for processing the weights of CRs and their relationship to PIECs, respectively. This leads to the identification of optimal PIECs that meet users' emotions, guiding the final NPD.

2. This study assists SMEs in quickly obtaining a large number of sketch alternatives that conform to PI elements and a complete and accurate understanding of user emotional needs, even when they lack the resources of senior designers and design experts, for refining the final design. This method not only meets the needs of enterprises for shaping brand influence but also effectively captures the minds of consumers, promoting the sustainable development of the enterprise.
3. The study explores the introduction of Midjourney into the real field of product design, combined with SG, to generate a large number of PI concept sketches. This process not only helps designers quickly obtain a multitude of ideas but also completely replaces the complex, inefficient, and highly limited human-driven creative phase, effectively enhancing the efficiency of ideation and significantly reducing the difficulty of the design process.

Nevertheless, this study has some limitations and needs to be followed up.

1. AI-powered tools in the field of product design tend to have a high degree of randomness during the creative process. Even with the use of multiple prompt words to constrain the output, designers still need to manually process the pre-trained image sources, which reduces work efficiency. Future research could explore the integration of different AIGC tools with the methods of this study to enhance the efficiency of concept sketch generation.
2. This study only investigates the formal design of PI, while Color, Material, and Finishing (CMF) are also elements of the PI design system. Future research should include these aspects to enhance the comprehensiveness of this research method in the PI system design.
3. In empirical studies, only the side view is considered, while in reality, the product exists in 3D space. Therefore, it is necessary to conduct further research on other perspectives.

6. Conclusions

This paper proposes a PI design method integrating SG, KE, and grey number theory. This method quickly generates a large number of conceptual sketches that meet the PI elements by combining SG and Midjourney, and obtains the grey weights of FCRs through online crawling of consumer reviews and G-AHP. It uses G-QFD to screen out the PIECs that best meet user emotions.

This paper selects a Chinese brand of female E-moped as an example for practice, analyzing 25 of its products to create 63 initial concept sketches with the Midjourney platform and SG method. It also mines requirements from JD and Amazon, analyzing 6,181 user reviews to identify 10 Kansei words. These words are then refined through focus groups and expert interviews to determine the weight of Kansei needs. MA and QFD methods are used to deconstruct 48 sketches, identifying 27 PIECs across 3 categories. Ultimately, the top 3 sketches that best meet the CRs are selected, and professional designers are invited to finalize the design solutions. By conducting an online questionnaire survey with 50 users, the results show that this method can effectively balance the impact of PI elements and user emotional needs on product form design. Hence, it is evident that the method possesses significant educational value in the realm of AI-powered product design, offering effective guidance for the design of diverse product forms.

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